

Antimicrobial resistance

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Key facts

- Antimicrobial resistance (AMR) occurs when bacteria, viruses, fungi and parasites do not respond to antimicrobial medicines. As a result, infections become difficult or impossible to treat, increasing the risk of disease spread, severe illness, disability and death.
- AMR is a major global health threat. Bacterial AMR was estimated to be associated with more than 4.7 million deaths globally in 2021 (1).
- The misuse and overuse of antimicrobial are driving the development and spread of drug-resistant pathogens.
- Lack of adequate access to appropriate new and existing vaccines, diagnostics and medicines also contribute to the crisis.
- Approximately 1 in 6 laboratory-confirmed bacterial infections worldwide were resistant to antibiotics in 2023.
- The world faces an antimicrobial research and development crisis, with few new medicines in the pipeline.

Overview

Antimicrobials – including antibiotics, antivirals, antifungals and antiparasitics – are medicines used to prevent and treat infectious diseases in people, animals and plants.

AMR occurs when bacteria, viruses, fungi and parasites do not respond to these medicines. As a result of drug resistance, infections become difficult or impossible to treat, increasing the risk of disease spread, severe illness, disability and death.

A global concern

AMR affects health, health-care systems and society at large.

Health impacts: AMR is associated with higher morbidity and mortality, and it narrows treatment options for patients with infections. It raises the risk of complications of medical procedures such as surgeries.

Health-care system impacts: AMR drives health-care costs, largely through increased hospital stay and higher need for intensive care. It also increases reliance on second-line antibiotics and additional diagnostic testing.

Societal impacts: AMR disproportionately affects vulnerable populations, reduces human productivity and erodes public trust in medicine. In addition, drug-resistant infections impact the health of animals and plants and reduce productivity in farms, threatening food security.

With the current level of action, total cost to treat resistant bacterial infections alone is predicted to reach US\$ 412 billion annually up to 2035, with additional productivity losses of US\$ 443 billion per year (2).

Key factors for the emergence and spread of AMR include the inappropriate use of antibiotics; poor access to clean water, sanitation and hygiene (WASH); inadequate infection prevention and control in households, health-care facilities and farms; limited access to vaccines, diagnostics and medicines; and lack of awareness and enforcement of relevant legislation. Too few new antibiotics are being developed to replace those that no longer work, leaving fewer treatment options for people with drug-resistant infections.

AMR affects all countries, regardless of income, and spreads across borders. People living in low-resource settings and vulnerable populations, such as the very young and very old, are especially impacted.

What is the present situation?

Antibiotic use

The misuse and overuse of antibiotics, for example taking them when they are not needed or taking the wrong type, remains one of the main drivers of AMR. In 2022, so-called access antibiotics – those recommended as first-choice treatments for most common infections – accounted for only about half (53%) of global human antibiotic use. So-called watch antibiotics, which carry a higher risk of resistance, made up roughly 45% of use and

exceeded 70% of total use in nearly one-third of countries. These figures fall short of the [global target](#) which calls for at least 70% of antibiotic use to come from the access group by 2030.

Drug resistance in bacteria

[One in six laboratory-confirmed bacterial infections](#) causing common infections in people worldwide in 2023 were resistant to antibiotic treatments. Between 2018 and 2023, antibiotic resistance rose in over 40% of the pathogen-antibiotic combinations monitored with an average annual increase of 5–15%. WHO estimates that antibiotic resistance is highest in the WHO South-East Asia and Eastern Mediterranean Regions, where 1 in 3 reported infections were resistant. Resistance is also more common and worsening in places where health systems lack capacity to diagnose or treat bacterial pathogens. Drug-resistant Gram-negative bacteria are becoming more dangerous worldwide, with the greatest burden falling on countries least equipped to respond. Among these, *E. coli* and *K. pneumoniae* are the leading drug-resistant Gram-negative bacteria found in bloodstream infections. These are among the most severe bacterial infections that often result in sepsis, organ failure and death.

In [gonorrhoea](#), a sexually transmitted infection, ciprofloxacin resistance is very high (75%), and ceftriaxone resistance – though still low globally – is rising in some regions, threatening the effectiveness of first-choice treatment.

Multidrug-resistant tuberculosis (MDR-TB) is a form of TB caused by bacteria that do not respond to isoniazid and rifampicin, the two most effective first-line TB drugs. MDR-TB is treatable and curable by using other drugs, which tend to be more expensive and toxic. The estimated proportion of people with a first episode of TB who had drug-resistant TB was [3.2% in 2024](#), while the proportion among those with a previous history of TB treatment was much higher, at 16%.

Drug resistance in fungi

Fungi can cause a wide range of infections, from [candidiasis](#) (thrush) of the mouth or genital area, and ringworm (dermatophytosis) of the skin, hair and nails, to severe and life-threatening infections such as invasive candidiasis or aspergillosis of the lungs. These infections disproportionately impact severely ill patients and those with weakened immune systems, including individuals undergoing cancer chemotherapy, organ transplantation and those living with HIV.

Fungal infections are increasingly difficult to treat due to AMR. The emergence and spread of multi-drug resistant *Candidozyma auris* (formerly *Candida auris*), an invasive fungal infection, is of particular concern. Drug-resistant *Aspergillus* is becoming harder to treat as many strains show resistance to azole antifungals, drugs widely used in both medicine and agriculture.

Drug resistance in viruses

Over the past decade, the use of antiretroviral therapy (ART) has saved the lives of tens of millions of people living with HIV. However, increased use of HIV medicines has been accompanied by the emergence of [HIV drug resistance](#). HIV drug resistance can lead to increased HIV infections and deaths.

Beyond HIV, antiviral resistance in influenza viruses is a major public health concern and is closely monitored by WHO through [global surveillance networks](#). WHO also recognizes emerging resistance challenges [affecting hepatitis B and C treatment programmes](#).

Drug resistance in parasites

The emergence of [drug-resistant parasites is a major threat to malaria control](#). Artemisinin-based combination therapies (ACTs) are the recommended first-line treatment for uncomplicated *Plasmodium falciparum* malaria. Emergence of partial artemisinin resistance makes selecting the right treatment more challenging and requires close monitoring.

The emergence of drug resistance against medicines for neglected tropical diseases (NTDs), such as [human African trypanosomiasis](#) and [leishmaniasis](#), is a significant threat to NTD programmes. Resistance has been reported in several key medicines, such as melarsoprol, pentavalent antimonials and miltefosine.

WHO response

WHO leads global efforts to prevent and control AMR in humans, as part of the [Global action plan on antimicrobial resistance \(GAP-AMR\) 2026–2036](#). The GAP-AMR aims to preserve the ability to treat human, animal and plant infections by expanding equitable access to and appropriate use of effective antimicrobials and reducing infections through a One Health approach.

By 2030, the plan aims to enable the attainment of the [2024 UN General Assembly target](#) of 10% reduction in bacterial AMR-associated deaths in humans, while also reducing antimicrobial use in agrifood systems, and minimizing environmental pollution from

resistant microbes and antimicrobial residues.

WHO works closely with the Food and Agriculture Organization (FAO) of the United Nations, the United Nations Environment Programme (UNEP) and the World Organisation for Animal Health (WOAH), together called [the Quadripartite](#), in a coordinated [One Health](#) approach to achieve the goals and objectives of the GAP-AMR.

GAP-AMR is operationalized through national action plans (NAPs), and a separate operational guidance and a monitoring and evaluation framework. The progress of NAPs is monitored through the Tracking AMR Country Self- Assessment Surveys ([TrACSS](#)) undertaken annually.

To achieve the GAP-AMR's goal, there are six interconnected strategic objectives:

1. strengthen awareness and promote appropriate social and behavioural change to reduce antimicrobial resistance risks across all sectors;
2. strengthen surveillance systems and laboratory networks to inform effective, evidence-driven antimicrobial resistance policies and actions;
3. intensify infection prevention across all sectors to reduce the burden of infectious diseases and the need for antimicrobials;
4. ensure equitable access, appropriate use and safe disposal of antimicrobials, diagnostics and other health products across sectors;
5. accelerate antimicrobial resistance research and innovation across all sectors; and
6. strengthen multisectoral governance, sustainable financing and accountability for a coordinated antimicrobial resistance response across all sectors and at all levels.

The GAP-AMR is complemented by the [WHO strategic and operational priorities to address drug-resistant bacterial infections in the human health sector, 2025–2035](#), to strengthen the sector-specific response to AMR.

References

1. GBD 2021 Antimicrobial Resistance Collaborators. Global burden of bacterial antimicrobial resistance 1990-2021: a systematic analysis with forecasts to 2050. Lancet. 2024 Sep 28;404(10459):1199-1226.
2. GLG report: Towards specific commitments and action in the response to antimicrobial resistance (2024). <https://www.amrleaders.org/resources/m/item/glg-report>

- [World Antimicrobial Awareness Week](#)